Sky Extension Activities

Five hands-on activities: a shadow stick, a sky in a jar, a pinhole Sun viewer, cloud watching and star stories

SAFETY AND CARE

- **Never look straight at the Sun**, not with sunglasses, a phone camera, binoculars or a telescope. In the pinhole activity children look at the **picture on paper**, never at the Sun.
- Do outdoor work early in the morning or late in the afternoon, in hats, with water.
- Use blunt pencils and safe scissors. Volunteers make the pinholes.

Activity 1: the shadow stick

MATERIALS

A straight stick about 30 cm long; clay or a lump of soil to hold it; a sheet of paper or a flat piece of ground; chalk or pencil; a ruler; a watch.

1. Put the stick **upright** in the clay in a sunny spot where the ground is flat. Do not move it again.
2. Every hour, mark the **tip of the shadow** with chalk, and write the time beside it.
3. Measure the length of the shadow each time and write it in the table.
4. Look at the marks. The shadow turns **in a curve** and gets shorter, then longer. The **shortest shadow** is at about noon, and it points **north** (in India, for most of the year). A line from the base through the shortest shadow is a north-south line.

KEY IDEA

The Sun seems to move because the Earth spins. The shadow tip draws the Sun's path for us, in the opposite direction.

NOTE

If clouds hide the Sun for a while, just skip that hour and continue. If you only have one class period, do three marks, 20 minutes apart, and still see the shadow move and change length.

Activity 2: a sky in a jar

MATERIALS
A clear glass or jar; water; a few drops of milk; a torch (phone light); a dark room or a dark cloth.

1. Fill the jar with water. Stir in just **three or four drops** of milk, so the water is only slightly cloudy.
2. In the dark, shine the torch **from the side**. Look at the jar from the side: the water looks faintly **blue**.
3. Now look **through** the jar at the torch, from the other end: the light looks **yellow-orange or red**.

The milk drops scatter blue light sideways, just as air does. Light that goes straight through has lost its blue and looks red: like the Sun at sunset. Ask: "Where in the jar is the daytime sky? Where is the sunset?"

TIP
If you add too much milk the water goes white and the effect is lost. Start with less; you can always add a drop.

Activity 3: a pinhole Sun viewer

MATERIALS
Two sheets of white card or paper; aluminium foil; tape; a pin. (A volunteer makes the hole.)

1. Cut a small square hole in one card and tape foil over it. Prick a **tiny hole** in the foil with the pin.
2. Stand with your **back to the Sun**. Hold the card with the hole so sunlight passes through it, and hold the second card as a screen, about an arm's length behind it.
3. On the screen you will see a **small round picture of the Sun**. Move the screen nearer and farther and watch the picture change.

SAFETY AND CARE
Look only at the screen. Never turn round and look at the Sun through the hole.

Light travels in straight lines. That is why the picture is round, like the Sun, and upside down in a real pinhole camera. When there is a partial solar eclipse, this viewer shows the Sun as a crescent, and it is a safe way to watch.

Activity 4: cloud watching

Lie on the grass (or sit) and look up for five minutes. Ask: "What shapes can you see?" Then name three kinds of cloud:

Cloud	Looks like	What it often means
Cumulus (তুলোমেঘ)	heaps of white cotton	fair weather
Stratus (স্তরমেঘ)	a flat grey sheet	grey day, maybe drizzle
Cirrus (পক্ষমেঘ)	thin high feathers	weather may change soon

Then draw the sky on a weather diary for a week: sky colour, clouds, wind, rain.

Activity 5: star stories

Stars make patterns that people have named in every culture. On a clear, dark evening (away from bright lights), look for these. Always check with a volunteer or a star app first, as the sky changes with the season.

Name	What to look for	Story or use
Saptarshi (সপ্তর্ষি)	Seven bright stars shaped like a big ladle or kite (the Big Dipper)	The seven sages. Its two end stars point to the Pole Star.
Dhruvatara (ধ্রুবতারা)	A fairly faint star that hardly moves, in the north, low over Kolkata	It tells us north at night.
Kalpurush (কালপুরুষ)	Three stars in a row (the belt) with a bright star on each side (Orion), in the winter evening sky	The hunter. Easy to find.
Shukatara (শুকতারা)	A very bright, steady "star" low in the east before sunrise (or west after sunset)	This is actually the planet Venus, and it does not twinkle.

NOTE
The stars and constellations visible depend on the season and the hour. Saptarshi is high in the northern sky on spring evenings and very low (or hidden) on autumn evenings. If a child cannot find it, say: "Try again another month."

TRY IT
Ask each group to invent their own star story about a pattern of stars they draw, and tell it to the class. Which animal, person or object do they see?

Student sheets

Name: _____	Date: _____

Shadow stick record

সময়	ছায়ার দৈর্ঘ্য (সেমি)	ছায়া কোন দিকে পড়েছে?

সবচেয়ে ছোট ছায়া ছিল _____ সময়ে। তখন ছায়া _____ দিকে ছিল।
The shortest shadow was at _____. Then the shadow pointed _____.

Sky in a jar

পাশ থেকে দেখলে জলের রং _____। ভেতর দিয়ে টর্চের আলো দেখলে রং _____।
Seen from the side the water looked _____. The torch seen through it looked _____.

Weather diary

এক সপ্তাহ রোজ আকাশ দেখে আঁকো বা লেখো।
Draw or write about the sky every day for a week.

দিন	আকাশের রং	মেঘ কেমন?	হাওয়া / বৃষ্টি
১			
২			
৩			
৪			
৫			
৬			
৭			

My star story

তারার নকশাটা এঁকে তার একটা নাম দাও ও গল্প লেখো।

Draw the pattern of stars, give it a name and write its story.